

A

Host Restriction Mechanisms Linked to Enriched Motifs

CpG → ZAP / OAS3

ZAP binds CpG-enriched viral RNA; OAS3/RNaseL provides complementary restriction (independent but functionally overlapping pathways).

No CpG-enriched motifs in top 10

UpA → ZAP / OAS3

ZAP also binds UpA-enriched RNA; OAS3/RNaseL cleaves at UpA/UU sites. Mechanism of attenuation remains uncharacterized.

6/10 motifs contain UpA

Composition Bias

Purine-rich, GC-rich, AU-rich patterns may affect RNA structure and stability.

3/10 motifs purine/GC-rich

B

Top 10 Discriminative 6-mer Motifs

Motif	$\log_2(E)$	Cons. ARB	Cons. ISFV	Peak Pos.	Biological Annotation
AUUAGG	3.22	0.60	0.21	0.78	UpA×1
CCCUUU	2.73	0.90	0.37	0.38	—
AGAGAG	2.59	1.00	0.94	0.78	Purine-rich
UACUCC	2.57	0.75	0.33	0.08	UpA×1
UAGUUA	2.51	0.42	0.20	0.88	UpA×2; AU-rich
AGAGGG	2.43	1.00	0.81	0.68	Purine-rich
CUAAUA	2.36	0.51	0.41	0.93	UpA×2
GGCCUC	2.33	0.91	0.53	0.33	GC-rich
GAAUAG	2.31	0.83	0.57	0.18	UpA×1
AUAGAA	2.24	0.86	0.49	0.73	UpA×1